

ARSHIN SIKKA

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EDUCATION

National University of Singapore

Aug 2023 - May 2027

Bachelor of Computing in Computer Science

- Artificial Intelligence Focus Area - Certificate of Distinction
- NUS Overseas Colleges — selected for the entrepreneurship programme, currently placed at Garuda Robotics
- Relevant Coursework: AI Planning & Decision Making, Computer Vision, NLP, Intro to AI & ML, Probability & Statistics, Linear Algebra, Data Structures & Algorithms, Software Engineering
- Teaching Assistant, AI Ethics and Intro to AI & ML — Taught weekly tutorials for ~50 undergraduates across two courses

SKILLS

- Programming Languages: Python, Java, TypeScript, JavaScript, SQL, HTML/CSS
- AI Frameworks & Orchestration: LangChain, LangGraph, MCP, RAG pipelines, multi-agent orchestration
- LLM Evaluation & Safety: Langfuse, guardrails, adversarial testing, penetration testing
- Machine Learning & Data: PyTorch, TensorFlow, scikit-learn, LoRA fine-tuning, pandas, NumPy, Monte Carlo simulation
- Backend & Databases: FastAPI, Node.js, Spring Boot, REST APIs, PostgreSQL, Supabase, Prisma, FAISS, pgvector
- Infrastructure & Testing: Docker, Fly.io, CI/CD, Git/GitHub, Bitbucket, pytest, JUnit

WORK EXPERIENCE

AI Engineer Intern, Garuda Robotics, Singapore

Jul 2026 - Present

- Architecting an agentic AI system end to end (conversational client, MCP gateway, MCP server) that turns a flight planner's plain-language brief into a validated drone flight plan.
- Designing the agent's questioning flow so a planner answers 3 questions instead of specifying 20 flight parameters.
- Building the guardrails that make an out-of-bounds flight plan impossible to generate.

Data Science Intern, SP Digital (Data & AI), Singapore

Jan 2026 - Jun 2026

- Engineered a context-aware guardrail layer for a production LLM assistant serving 100+ grid fieldworkers, closing the findings of an external penetration test.
- Built an automated adversarial evaluation pipeline using Langfuse covering 400+ attack prompts across multiple regression datasets, cutting unsafe model responses by ~60% across internal deployments.

AI Engineer Intern, KPMG, Gurugram, India

May 2025 - Aug 2025

- Designed an agentic RAG system using LangChain and Azure OpenAI embeddings over a FAISS vector store, replacing manual SharePoint search across hundreds of consulting documents.
- Augmented the retrieval engine with custom document-parsing layers to cite source PDFs and extract structured Excel data, cutting consultant knowledge lookup time 40-50% across a 12-person team.

SDE Intern, AlygnAI, California, USA (Remote)

Jun 2025 - Aug 2025

- Benchmarked LoRA fine-tuning against RAG for domain adaptation, producing the performance, cost and latency profiles behind the founding team's architecture choice.
- Rebuilt a no-code AI prototype into a production FastAPI backend with JWT authentication, bcrypt hashing and 2FA.

PROJECTS

Slipstream, PSA Code Sprint 2.0 ([GitHub](#))

Aug 2026 - Sep 2026

- Built an agentic system that reroutes containers stranded by a late ship, letting the model read contracts while a CP-SAT solver decides every number.
- Scored every decision on four criteria, automating 70% of routing calls and escalating the rest for human review.

Socratic Digital Twin, NUS Faculty of Dentistry

Jun 2026 – Present

- Encoding senior specialists' clinical reasoning into a 10-node LangGraph pipeline over pgvector, building an AI tutor that guides orthodontics residents through a case instead of giving them the answer.
- Routing between OpenAI, Claude and DeepSeek through a model-agnostic gateway, with a guardrail layer that regenerates any response breaking the questioning-only constraint.

Singapore Society Simulation, NUS Odyssey ([GitHub](#))**May 2026 - Aug 2026**

- Developed a multi-agent policy simulation platform across 100 NVIDIA Nemotron Singapore personas, supervised by Prof See-Kiong Ng and selected for public academic showcase at NUS School of Computing.
- Measured belief drift across deliberation rounds, benchmarking simulated opinion shifts against real Reddit discourse on the same policy questions.

Lecture AI, Co-Founder, Singapore ([Website](#))**Jul 2025 - Mar 2026**

- Shipped an end-to-end AI pipeline converting university lecture recordings into structured study notes, chaining audio preprocessing, Whisper transcription, and a two-pass correction and summarisation step at under \$1 per lecture.

LEADERSHIP

Director of Human Resources, NUS Student Union (NUSSU)**Nov 2024 - Present**

- Designed onboarding and feedback systems for NUS's apex student body, standardising how 100+ student leaders are inducted and reviewed across cross-functional teams.

Professional Player & Team Captain, Chess**Apr 2015 - Present**

- Represented India at the 2019 Commonwealth Chess Championship and placed runner-up in a FIDE-rated tournament of 693 players; captained NUS to Inter-Faculty Games victory.